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A sign audit of a Hessian term absent from the standard MOND equation: action-consistent virial work and conditional Jeans tests in three ultra-faint dwarfs

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Abstract

This note recasts the UDCT v7.01–v7.06 calculations as a verification paper. Standard MOND-family field theories (AQUAL, QUMOND) employ a positive interpolating function μ or ν . Their weak-field equations do not contain the extra Hessian-sourced tensor P_{ij} tested in this UDCT candidate [3,4]. The object under audit is therefore not standard AQUAL or QUMOND. It is one unregistered, smooth, Hessian-augmented field law whose multiplier ν and tidal tensor P_{ij} are obtained from a single action integral Q . Where the density-weighted virial integral is negative, that sign is not produced by a negative interpolating function. It is produced because the action-derived Hessian contribution exceeds the positive leading term. A previously preregistered ultra-faint-dwarf velocity proxy (O8d / D8b) matched 0/22 objects within a factor of two; that historical gate is not reopened. After replacing a circular-orbit external acceleration with a visible Milky Way baryon field, the same factor-two proxy still fails 0/22. For the later smooth action, a density-weighted virial integral is negative in Leo IV, Pegasus III and Hydrus I at the original transition scale for switch exponents $p = 1\text{--}8$ and for both a Plummer profile and a tapered cusp. Global work changes sign if the adopted Galactic acceleration and Hessian are scaled together to only 0.32%–10.56% of their baseline strength. Raising the transition scale by a factor of about 440–1,010 restores positive work, but the field law then approaches the QUMOND form. Even on that positive-work branch, a conditional spherical-isotropic Jeans calculation gives projected dispersions inside one effective radius of only $0.55\text{--}1.19\text{ km s}^{-1}$, against archived values $2.7\text{--}5.4\text{ km s}^{-1}$. One positive-work Pegasus III example further develops a negative radial Jeans pressure under the stated spherical, isotropic, static and outer-pressure assumptions. The result is a diagnostic trilemma for this candidate, not a general impossibility theorem for modified gravity or for bidirectional projection, and not a sign theorem for AQUAL or QUMOND.

Keywords: modified Newtonian dynamics; Hessian tensor; external field effect; ultra-faint dwarfs; virial theorem; Jeans equation; tidal tensor; audit methods

Status. Post-outcome research record. No new constitutive law is registered. Numerical values are taken from UDCT v7.01–v7.06. This file is a wording revision of the 29 September 2026 verification draft; no new calculations were performed. Informal large-language-model commentary is acknowledged only as prior discussion, not as a source of results.

1. Purpose and limits of this verification note

Modified Newtonian Dynamics (MOND) was introduced to account for galactic mass discrepancies without dark-matter particles [1,2]. Complete non-relativistic field theories exist: the aquadratic Lagrangian theory (AQUAL) [3] and the quasi-linear formulation (QUMOND) [4]. Both employ an interpolating function whose Newtonian and deep-MOND limits are fixed by the paradigm, while the detailed shape is not derived from a parent theory [5,6]. Monotonicity of $x\mu(x)$ is required for a one-to-one map between the true and Newtonian fields and for a convex Lagrangian with standard conservation laws [5,7]. Those constraints are therefore not merely a fit to rotation curves. They are well-posedness conditions. They are also, as MOND authors themselves note, waiting for a deeper derivation [6,8].

A separate question is what happens if the leading multiplier remains a standard positive interpolator, but a higher-order tidal or Hessian term is added and then differentiated consistently with the same action. That is the object audited here. The candidate is the unregistered smooth-action family studied in UDCT v7.02–v7.06, in which a tidal invariant built from the Newtonian Hessian sources an extra tensor P_{ij} . The present paper does not propose this family as a replacement for MOND or for dark matter. It asks whether the family survives the elementary equilibrium tests that any static collisionless model must pass in the ultra-faint dwarfs of the Milky Way.

Three statements are kept separate throughout, following the v7.01 reading rule [9]:

- (A) The preregistered O8d / D8b velocity-dispersion proxy failed (0/22 versus a required 11/22).
- (B) Corrected but still simplified proxies and later unregistered diagnostics also fail for the stated profiles and Galactic field.
- (C) The complete action-derived field has not been solved as a self-consistent three-dimensional equilibrium with an observed stellar distribution function. A negative toy-model integral is an obstruction inside that toy model, not a measured significance level and not a theorem about every modified-gravity completion.

The scientific product offered for deposition is therefore a verification protocol: action-consistent differentiation of the switch, joint scaling of the external acceleration and Hessian, band-wise integration by parts of the derivative work, and a conditional Jeans test that can reveal a negative radial pressure even when the global virial integral is positive. The protocol is illustrated on one family. It is intended to be reusable on any later Hessian- or curvature-augmented extension.

2. Historical O8d gate and the independent v7.01 audit

The v5.60 pre-registration fixed the constitutive law, estimator, decision rule and test order. The v5.61 registered proxy then failed option D8b: 0 of 22 Milky Way ultra-faint dwarfs lay within a factor of two of the archived line-of-sight dispersion, against a pass rule of at least 11/22 [9]. That verdict is historical and is not re-registered by any later note.

v7.01 identified a model-input inconsistency in that proxy. The external acceleration fed to D8b was a physical circular-orbit estimate v^2/D with a heliocentric distance D , whereas the frozen field law requires the Newtonian field of visible baryons evaluated at the Galactocentric position [9]. Replacing the input by the stellar-plus-gas field of the McMillan (2017) baryonic mass model [10], correcting distances to Galactocentric radii, and vector-averaging the leading field term still gave 0/22. The median predicted-to-observed ratio was about 0.09. Eight of 22 objects then had compactness switch $S < 0.5$, so the earlier claim that screening was inactive in every system does not survive the correction [9].

Evaluating the action-derived derivative term on two idealized three-dimensional dwarf profiles produced a large opposing monopole in 16/22 cases. v7.02 showed that this term is not an artefact of the non-differentiable spectral invariant $\tau = \max |\text{eigenvalue}(H)|$: a smooth, point-mass-value-matched replacement still yields 16/22 outward mean radial forces once the Galactic field is restored, and zero of 22 when that field is removed [11]. v7.03–v7.04 moved from proxy half-mass radii to Local Volume Database projected half-light radii [12] and from a point Galaxy to an integrated visible disk. The outward-mean count at catalog radii became 17/22 for stellar $M/L = 2$ [13]. These figures are diagnostics of the later unregistered action. They do not repair the frozen O8d gate.

3. Field law under audit

From v7.05 onward the tested smooth invariant and source equation are [14,15]

$$\tau = [(2/3) \text{tr}(H^2)]^{1/2}, A_{ij} = 2H_{ij} / (3\tau), x = |\nabla\Phi_N| / a_0, X = c^2\tau / (a_0^2 x^2).$$

$$\begin{aligned}
S_p(X) &= 1 / [1 + (X/X_s)^p], \quad \nu = 1 + S_p(X) [\nu_c(x) - 1], \\
\nu_c(x) &= 1 + [\sqrt{(1 + 1/x)} - 1] / [1 + (x/490)^2], \\
Q(z, \tau) &= \int_0^z \nu(\sqrt{s}, c^2\tau / (a_0^2 s)) ds, \quad P_{ij} = a_0^2 Q_\tau A_{ij}, \\
\nabla^2 \Phi &= \nabla \cdot (\nu \nabla \Phi_N) - (1/2) \partial_i \partial_j P_{ij}.
\end{aligned}$$

Baseline parameters are $p = 4$, $X_s = 4 \times 10^9$ and $a_0 = 1.082401 \times 10^{-10} \text{ m s}^{-2}$. The visible McMillan disks plus a point bulge of 8.62×10^9 solar masses supply one mutually consistent external acceleration E and Hessian H_{MW} at each object; the integrated baryonic mass in that quadrature is 6.621×10^{10} solar masses [10,14]. Locally, $\nabla \Phi_N = g_*(r) \mathbf{n} + E + r H_{MW} \mathbf{n}$. Stellar mass-to-light ratio is 2 in solar units. Discarding P_{ij} would change the theory under test. Rescaling P while holding ν fixed would break action consistency.

Two stellar profiles are used, both normalized to the same projected half-light radius R_e from the Local Volume Database [12,14]. The Plummer model has scale $b = R_e$. The alternative is a Hernquist-like cusp with a smooth taper $\exp[-(r/(10R_e))^4]$, with inner scale chosen so that half the projected mass lies inside R_e . Catalog inputs at the v7.05 values are Leo IV: $L = 8472 L_\square$, $R_e = 101.11 \text{ pc}$, $D_{GC} = 151.93 \text{ kpc}$; Pegasus III: $3733 L_\square$, 82.12 pc , 212.84 kpc ; Hydrus I: $6546 L_\square$, 52.42 pc , 25.68 kpc . Hydrus I is classified as an LMC satellite in that catalog; the LMC potential is absent from the present field [12,14].

4. Verification protocol

4.1 Density-weighted virial work

For a stationary isolated collisionless system in a conservative scalar field, with vanishing surface stresses, the gravitational work

$$V = \int \rho_*(r) r^3 \langle g_r(r) \rangle dr$$

must be positive if it is to balance $2K$ [14]. The angular-mean radial acceleration includes both the ν -weighted Newtonian term and the action derivative of P . We report V/V_N , where $V_N > 0$ is the same integral built from the stellar Newtonian field of the same profile. Negative V is an obstruction to that assumed equilibrium. It is not itself a dispersion measurement.

4.2 Joint scaling of the external field

Set $E \rightarrow \alpha E$ and $H_{MW} \rightarrow \alpha H_{MW}$ at fixed stellar profile, luminosity and action. $\alpha = 0$ is essentially Newtonian in this calculation ($V/V_N \approx 1$). $\alpha = 1$ recovers the adopted visible-Galaxy field. The crossing α_{crit} is a diagnostic of the unregistered action, not an inference of the true Milky Way mass [15].

4.3 Action-consistent switch scans

For arbitrary exponent p the τ -derivative of Q is evaluated from the same integral that defines ν . Changing p or X_s therefore changes both the multiplier and the tensor P . A scan that varied only one of them would not be a variation of the action. The identity used in v7.06 is [15]

$$W \equiv a_0^2 Q_\tau = -(p c^2 / X) \int_0^1 S_p(X/t) [1 - S_p(X/t)] [\nu_c(x\sqrt{t}) - 1] dt.$$

4.4 Conditional spherical-isotropic Jeans test

On selected positive-work branches we integrate

$$d[\rho_* \sigma_r^2]/dr = -\rho_* \langle g_r \rangle, \quad \rho_* \sigma_r^2(r) = \int_r^{R_{\text{out}}} \rho_*(s) \langle g_r(s) \rangle ds,$$

with zero outer radial pressure. The Galaxy remains a local Taylor field. This is not a self-consistent three-dimensional Jeans solution in the nonaxisymmetric potential. Isotropic variance is then averaged in a projected cylindrical aperture of radius R_e [15]. A negative σ_r^2 anywhere in the stellar body is a sharper obstruction than a positive global V is a success.

5. Results

5.1 Baseline negative virial work (v7.05)

At $M/L = 2$ and catalog R_e , the normalized work is negative for every object and both profiles [14]:

Object	Plummer V/V_N	Tapered cusp V/V_N	Plummer $\langle g/g_* \rangle$	Taper $\langle g/g_* \rangle$
Leo IV	−683.84	−969.06	−10.20	−11.32
Pegasus III	−810.90	−1489.62	+0.88	+0.87
Hydrus I	−415.42	−299.67	−483.44	−462.21

Table 1. Baseline density-weighted virial work and angular-mean force at the profile 3D half-mass radius, from v7.05. Pegasus III shows why a locally positive mean force does not certify equilibrium: its global integral is still negative.

All 27 illustrative corners per profile ($M/L = 1, 2, 4$ and R_e scaled by 0.7, 1.0, 1.3) remain negative [14]. Setting both E and H_{MW} to zero returns $V/V_N = 1$. A uniform external gradient alone already drives a large negative response; adding the consistent tide moderates the number but does not restore a positive sign.

5.2 External-field threshold for a sign change (v7.06)

Object	Plummer α_{crit}	Tapered-cusp α_{crit}	Interpretation
Leo IV	0.01903	0.06425	1.90% / 6.43% of adopted E, H
Pegasus III	0.02426	0.10556	2.43% / 10.56%
Hydrus I	0.00320	0.00520	0.32% / 0.52%

Table 2. Joint scaling of the visible-Galaxy acceleration and Hessian. Work changes from positive to negative at a few percent of the adopted field. Varying the Plummer outer boundary from 40 to 160 half-mass radii shifts the crossing by at most about 0.03 percentage points [15].

If ν were held fixed and only P were multiplied by an artificial factor λ , the largest λ still compatible with nonnegative global work at $\alpha = 1$ would be 0.79%/0.54% (Leo IV), 0.37%/0.27% (Pegasus III) and 0.95%/1.18% (Hydrus I) for Plummer/tapered profiles. Those percentages measure the dominance of the derivative term. They are not candidate fits [15].

5.3 Where the negative derivative accumulates

Object	Dominant band	Plummer share	Tapered-cusp share
Leo IV	2–8 r_h	86.0%	89.3%
Pegasus III	2–8 r_h	87.4%	92.8%
Hydrus I	0.5–2 r_h	75.8%	67.5%

Table 3. Fraction of the negative action-derivative contribution, after integration by parts, coming from specified bands in units of each profile's own 3D half-mass radius r_h [15]. The obstruction is not generally located at a single half-mass-radius force sample.

5.4 Switch-exponent scan at the original transition scale

Object / profile	$p = 1$	$p = 4$ (baseline)	$p = 8$
Leo IV / Plummer	−605.5	−683.8	−685.1
Leo IV / taper	−959.2	−969.1	−958.4
Pegasus III / Plummer	−706.1	−810.9	−813.0
Pegasus III / taper	−1266.0	−1489.6	−1488.9
Hydrus I / Plummer	−722.6	−415.4	−399.1
Hydrus I / taper	−489.7	−299.7	−288.7

Table 4. Normalized virial work V/V_N at fixed $X_s = 4 \times 10^9$. Every intermediate exponent $p = 2, 3, 6$ was also negative [15]. A slope change alone does not restore positive work inside this tested family. The scan is finite; it is not a theorem about all smooth actions.

5.5 Moving the transition scale

Very small X_s suppresses the modification and returns a nearly Newtonian branch ($V/V_N \approx 1$). The original 4×10^9 lies inside a large negative-work interval. An upper-scale positive branch exists [15]:

Object / profile	High- X_s crossing	$\times$ baseline	V/V_N at 4×10^{13}	V/V_N at 4×10^{17}
Leo IV / Plummer	2.374×10^{12}	593 $\times$	12.318	14.013
Leo IV / taper	2.382×10^{12}	596 $\times$	12.208	13.895
Pegasus III / Plummer	3.891×10^{12}	973 $\times$	15.982	19.370
Pegasus III / taper	4.034×10^{12}	1008 $\times$	15.758	19.214
Hydrus I / Plummer	2.263×10^{12}	566 $\times$	3.525	3.994
Hydrus I / taper	1.756×10^{12}	439 $\times$	3.210	3.557

Table 5. First zero-crossing of V on the high- X_s branch and selected values deeper in that branch, at $p = 4$ [15]. Thresholds were located in $\log X_s$ and rechecked on a 24×36 angular, 192-node action grid. They are conditional thresholds, not optimized fits.

At $X_s = 4 \times 10^{17}$ the derivative contribution is very small and the source equation tends to $\nabla^2 \Phi = \nabla \cdot [\nu_c (|\nabla \Phi_N|/a_0) \nabla \Phi_N]$. That is the QUMOND equation form with the specified ν_c [4,15]. High- X_s behavior is therefore not evidence for a distinct measured bidirectional effect. Positive integrated work is only a necessary condition for the chosen stationary profile. It does not demonstrate a nonnegative distribution function, a 3D stationary solution, a match to measured stellar velocities, or compatibility with solar-system, rotation-curve or lensing gates.

5.6 Conditional Jeans dispersions on the positive-work branch

Object / profile	LOS rms at R_e , $X_s=4 \times 10^{13}$	LOS rms at R_e , $X_s=4 \times 10^{17}$	Archived observed
Leo IV / Plummer	1.067 km s ⁻¹	1.123 km s ⁻¹	3.3 km s ⁻¹
Leo IV / taper	1.145	1.190	3.3
Pegasus III / Plummer	0.909	0.977	5.4
Pegasus III / taper	0.979*	1.033	5.4
Hydrus I / Plummer	0.554	0.614	2.7
Hydrus I / taper	0.603	0.648	2.7

Table 6. Projected isotropic line-of-sight rms inside one effective radius on two high- X_s models, versus archived v5.61 dispersions used only as a scale reference [9,15]. *The Pegasus III tapered value at 4×10^{13} is not an equilibrium prediction (Section 5.7). Outer limits: $80 r_h$ (Plummer) or $40 r_h$ (taper).

A virial cross-check at $X_s = 4 \times 10^{13}$ gives global Plummer LOS rms values 0.934, 0.783 and 0.610 km s⁻¹ for Leo IV, Pegasus III and Hydrus I. Direct radial Jeans integration reproduces those global numbers to the printed precision [15]. The R_e aperture values are larger for the two more distant objects, but remain far below the archived figures under the stated M/L and profile assumptions.

5.7 Pegasus III: positive global work with negative local pressure

For the tapered Pegasus III model at $X_s = 4 \times 10^{13}$, $V/V_N \approx 15.758$ is positive. The angle-averaged inward force nevertheless becomes outward around $6\text{--}8 r_h$. With the stipulated zero-pressure outer boundary, the integrated isotropic radial pressure is negative over approximately $5.23\text{--}6.75 r_h$. At about $5.94 r_h$ the inferred σ_r^2 is ≈ -0.00623 (km s⁻¹)² [15]. No everywhere-nonnegative isotropic dispersion profile therefore exists under this spherical-average reduction.

Radial $\times$ angular $\times$ action grid	g_r/g_* at $6 r_h$	Minimum σ_r^2 in $4\text{--}9 r_h$
$500 \times 12 \times 20 \times 96$	-0.803	-0.006179 (km s ⁻¹) ²
$1000 \times 24 \times 36 \times 160$	-0.837	-0.006217
$1800 \times 32 \times 48 \times 256$	-0.894	-0.006229

Table 7. Grid refinement for the Pegasus III tapered model at $X_s = 4 \times 10^{13}$ [15]. The negative-pressure band is stable in sign. At $X_s = 4 \times 10^{17}$ this particular band disappears, but the projected dispersion remains too small (Table 6).

6. Discussion: the extra Hessian term and the origin of the negative sign

6.1 What is, and is not, claimed about AQUAL and QUMOND

Standard AQUAL and QUMOND use a positive interpolating function $\mu(x)$ or $\nu(y)$. Their weak-field source equations do not contain the extra tensor P_{ij} built from the Newtonian Hessian and then differentiated from the action integral Q [3,4]. The present candidate keeps a MOND-like positive multiplier and adds that tensor. This note does not evaluate the virial-work or Jeans sign of AQUAL or QUMOND on Leo IV, Pegasus III or Hydrus I, and it does not prove a general sign theorem for those theories.

Where the density-weighted virial integral of the candidate is negative, the interpolating function itself has not become negative. The leading ν -weighted field term remains a positive contribution in the decomposition used here. The negative sign appears because the action-derived Hessian contribution, after consistent differentiation of Q , exceeds that positive leading term. Sections 5.2–5.3 locate the same dominance in the joint external-field threshold and in the radial bands of the derivative work. Artificially rescaling P while holding ν fixed would break action consistency; it is reported only as a measure of how large the derivative term is, not as a physical fit [15].

AQUAL's well-posedness conditions on μ , including monotonicity of $x \mu(x)$ and convexity of the Lagrangian, remain the standard reasons those field theories are constructed as they are [3,5,7]. They are not used in this paper as a proof that AQUAL or QUMOND would return a positive virial integral or a nonnegative Jeans pressure under the same profiles, Galactic field and apertures. That comparison is listed as future work (Section 7, item F5).

The negative radial Jeans value in Section 5.7 is likewise conditional. Under the stipulated spherical, isotropic, static reduction, with vanishing outer radial pressure, one positive-work Pegasus III model has $\sigma_r^2 < 0$ on an outer radial interval. That is an incompatibility of this candidate with those assumptions. It is not a general statement about MOND interpolators, and it need not survive anisotropy, rotation, time dependence or a different outer boundary.

6.2 The diagnostic trilemma of the present candidate

The calculations narrow one candidate. They do not establish a general impossibility result for bidirectional projection or for modified gravity [15]. What they do establish, inside the stated profiles, Galactic Taylor field, and differentiable action family, is a trilemma.

Branch I — original transition scale. For $p = 1$ –8 at $X_s = 4 \times 10^9$, the density-weighted virial work is negative in all three objects. The derivative term dominates. Jointly reducing the adopted Galactic field to a few percent of its baseline strength is enough to flip the sign, which is a statement about the action, not a licence to refit the Milky Way.

Branch II — retuned transition scale. Raising X_s by a factor of about 440–1,010 restores $V > 0$, but the extra tensor dies and the source equation approaches QUMOND [4]. The bidirectional content that distinguished the candidate from existing MOND field theories is then no longer doing the work.

Branch III — positive V is not sufficient. On the high- X_s branch the conditional Jeans aperture dispersions remain a factor of several below the archived values. In one intermediate Pegasus III model the global integral is positive while $\sigma_r^2 < 0$ on an outer stellar shell. A global virial success can conceal a local Jeans failure.

The practical point is narrower. If one keeps a MOND-like positive multiplier but adds a Hessian-sourced tensor and differentiates the action all the way, positive ν does not by itself guarantee a positive virial integral. In this family the extra term can dominate. Restoring the sign

by shrinking the adopted Galactic field, by rescaling P alone, or by sending X_s toward the QUMOND limit, changes the candidate rather than confirming it.

Two further cautions follow from the earlier notes. First, 0/22 is a preregistered factor-two count, not a likelihood ratio, and the registered gate did not propagate velocity or mass-profile uncertainties [9]. Second, ultra-faint dwarfs are themselves difficult objects: membership, binaries, tides and non-equilibrium are live observational issues [16–19]. A toy spherical Jeans integral cannot settle those issues. It can only show that, under the assumptions used here, the candidate does not supply a static isotropic equilibrium with the archived scale of motions.

7. Future directions

The v7.06 record already listed the next gates [15]. They are restated here as a research programme, with the literature that makes each gate concrete.

F1. A new constitutive action with a bounded derivative. The immediate requirement is not another numerical value of X_s . It is an action whose tidal or gradient invariants are justified by a parent construction and whose variation cannot dump an unbounded opposing monopole into a dwarf. Generalized QUMOND with an additional length or frequency scale [20] and related multi-potential formulations are the relevant comparison class, because they change how tides enter without abandoning an action principle. Any replacement must be differentiated consistently. Independently rescaling P while holding v fixed is not a physical solution [15].

F2. Full nonaxisymmetric field before spherical Jeans. The present Galaxy is a local Taylor expansion. Leo IV already showed a mixed directional field in a finite-ball solve [13]. The next calculation should solve the complete source equation in the nonaxisymmetric potential with a controlled outer boundary, then reduce to kinematics. Spherical Jeans on an angle-averaged force is a screen, not a substitute.

F3. A nonnegative stellar distribution function, or an explicit tidal non-equilibrium model. Negative σ_r^2 under isotropy may or may not survive anisotropy, rotation or time dependence. The honest next step is either a nonnegative distribution function in the solved potential, or a dynamical model in which the dwarf is not assumed static. Observed tidal features in Milky Way satellites are themselves a live MOND-versus-CDM diagnostic [21,22] and must be treated as dynamics, not as a licence to ignore the virial sign.

F4. Include the LMC for Hydrus I. Hydrus I is an LMC satellite in the Local Volume Database [12]. The present field omits that potential. Any claim about Hydrus I that is intended for comparison with data must rebuild E and H_{MW} with the LMC included and then freeze the prediction.

F5. A like-for-like QUMOND / AQUAL control on the same three objects. To earn the phrase “MOND-family verification”, the same luminosities, radii, M/L values and apertures should be run through standard QUMOND with the external field effect [4,23–25]. Ultra-faint dwarfs are already a recognized tension for MOND under some treatments of the external field [19,26]. The point of the control is not to relitigate that literature. It is to show what, if anything, the extra P tensor changes once the comparison is fair.

F6. Do not reopen solar-system, rotation-curve or lensing gates by a dwarf-only retune. Interpolation-function families that fit galaxies can fail lunar laser ranging and Cassini tracking [5,27]. A scale that is raised by two to three orders of magnitude to save three dwarfs must be put back through those gates before it is called a theory. The same applies to cluster lensing and to the residual missing mass in clusters, which MOND has never fully removed [2,5].

F7. Public code, frozen predictions, then comparison. The series v7.01–v7.06 is a private research record. A deposit that aspires to verification should release the quadrature rules, the McMillan disk integration, the Q integral, the Jeans integrator and the exact object table. Predictions should be frozen before they are compared with matched-aperture velocities. No later note should be allowed to inherit the rhetorical status of the v5.61 registration without inheriting its decision rule.

8. Conclusion

UDCT v7.06, read together with the preceding audit notes, supports four conclusions and does not support a fifth.

First, the preregistered O8d velocity proxy failed, and a baryon-field correction of that proxy still fails 0/22. That historical fact stands.

Second, for the later unregistered smooth action, a density-weighted virial integral is negative in Leo IV, Pegasus III and Hydrus I at the original transition scale, for both stellar profiles tested, and for switch exponents 1 through 8. The negative contribution is carried by the action derivative of the Hessian term and is spatially extended, not confined to a single half-mass-radius sample.

Third, the only nearby repairs inside this family are diagnostic rather than physical: shrink the adopted Galactic field to a few percent of its baseline, artificially down-weight P while holding v fixed, or raise X_s until the extra tensor dies and the law resembles QUMOND. The last of these restores a positive global integral but still underpredicts the archived aperture dispersions, and it can leave a negative radial Jeans pressure in Pegasus III.

Fourth, the reusable content of the note is the protocol itself. Joint scaling of E and H , action-consistent variation of the switch, band-wise integration by parts, and a Jeans test that is allowed to return $\sigma_r^2 < 0$ even when $V > 0$, form a minimum stress test for any later theory that promotes a tidal or curvature tensor into the gravitational source equation.

What the record does not support is a general statement that AQUAL or QUMOND fail the same sign tests, or that every bidirectional or higher-order extension must fail. Those theories were not computed here under the same protocol. The present candidate adds an action-differentiated Hessian term, keeps a positive interpolator, and then encounters negative work and, under the stated Jeans assumptions, a local negative pressure. That is a result about this action family in three ultra-faint dwarfs under stated approximations. It is enough to reject the candidate as a static explanation of those objects. It is not enough to close modified gravity.

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Appendix A. Version trail used in this deposit

Note	Date	Role in this verification paper
v5.61 / v5.62	archived	Preregistered O8d / D8b gate: 0/22 vs required 11/22
v7.01	28 Sep 2026	Independent audit; baryon-field correction still 0/22; 16/22 opposing derivative
v7.02	28 Sep 2026	Smooth tidal invariant; outward means appear only with the Galactic field
v7.03	28 Sep 2026	Consistent tide + 3D force; 16/22 outward means; negative virial examples
v7.04	28 Sep 2026	LVDB radii; 17/22 outward at $M/L = 2$; Leo IV mixed directional field
v7.05	29 Sep 2026	Disk-derived E, H; $V/V_N < 0$ for both profiles in three dwarfs
v7.06	29 Sep 2026	α thresholds, p and X_s scans, Jeans test, Pegasus III $\sigma_r^2 < 0$

Table A1. Documents consolidated in this Zenodo record. Later notes do not reopen the frozen O8d decision rule and do not register a new constitutive law.

Numerical rules carried forward from v7.06: three-dimensional radial and angular integration; transformed action integral $u = t / (t + X/X_g)$; centered radial derivatives; outer-pressure-zero Jeans integral. The present PDF is a

post-outcome verification record, not a preregistered test.

End of the v7.06 verification paper. The frozen 0/22 gate and the lack of a demonstrated bidirectional back-reaction remain unchanged.